

risa: Basic functions

Joao Bosco Gusmao

2026-04-02

The *risa* package

The **risa** package, which stands for **RISk Assessment**, was designed to automate the application of the Habitat Risk Assessment (HRA) model, which is part of the *InVEST (Integrated Valuation of Ecosystem Services and Tradeoffs)* toolkit developed by the Natural Capital Project (Arkema et al., 2014; Sharp et al., 2020).

The package provides a suite of functions designed to automate the main steps required to apply the *ByRA* (Bycatch Risk Assessment) framework (Hines et al., 2020; Wilhelm et al., 2022). *ByRA* adapts the Habitat Risk Assessment (HRA) model to estimate bycatch risk by combining spatial overlap between fishing activities and marine megafauna with information on species vulnerability. In this framework, risk is estimated from two main components: *exposure*, which describes the degree to which a species is likely to interact with a given fishery, and *consequence*, which represents the expected severity of that interaction for the species or population. Thus, *ByRA* integrates spatial patterns of fisheries and species occurrence with criteria related to susceptibility, sensitivity, and potential population-level impacts. Although *ByRA* has been applied mainly to assess bycatch risk for marine mammals, the HRA implementation provided in **risa** is flexible and can accommodate different formulations or extensions of the HRA model, following the general approach implemented in the *InVEST* toolkit.

When fitting the HRA model, users can choose between multiplicative and Euclidean formulations of risk, apply optional decay functions to account for stressor influence buffers, and generate detailed summary statistics. In addition, **risa** provides flexibility to customize model settings, including the number of risk classes, buffer sizes for kernel density estimates and stressor zones, and the choice of coordinate system, either metric or decimal degrees.

Specifically, **risa** includes functions to generate kernel density maps of species occurrences and stressor intensities from either data frames or shapefiles, create overlap hotspot maps, and run the full HRA model for one or multiple species and stressors. It also includes functions for handling raster and vector maps, generating figures that summarize HRA results, exporting raster or vector outputs, and quickly implementing the *ByRA* framework from raw observation data, either for exploratory analysis or for formal applications.

Installation

Since **risa** is still in its *alpha* version, it cannot be installed from the CRAN repository. Therefore, **risa** must be installed from its GitHub repository:

<https://github.com/GusmaoJB/risa/>

The easiest way to install **risa** is to use the `install_github()` function from the **devtools** package. If **devtools** is not already installed, first install it with `install.packages("devtools")`, and then run the code below to install **risa**.

```
# Load devtools
library(devtools)

# install the risa package
```

```
devtools::install_github("GusmaoJB/risa")
```

```
# Load the risa package  
library(risa)
```

Performing ByRA

Since we depend on functions in the packages *terra* and *sf* to load and manipulate rasters and shapefiles, we will load these packages to perform our examples. First, let's keep our work files (rasters, shapefiles, dataframes) in a *data* folder inside our working directory. In this first example, let's suppose we have species and stressor distributions as occurrence points in CSV files:

```
# Load important packages  
library(sf)  
library(terra)  
  
# Loading species and stressor data  
spp_df <- read.csv("data/species_distr.csv")  
str_df <- read.csv("data/stressor_distr.csv")
```

Then we have a dataframe describing the occurrences of different species:

```
# Check the first rows  
head(spp_df)
```

##	long	lat	species	group_size
## 1	-4.4417028	-0.30863054	Dolphin	1
## 2	4.7315084	-0.27739997	Dolphin	1
## 3	-2.8702334	-0.14875624	Dolphin	2
## 4	-2.7600157	-0.08447276	Dolphin	3
## 5	-5.9929263	-0.12344416	Dolphin	4
## 6	-0.8168881	-0.24008144	Dolphin	7

And a data.frame with similar structure for stressors:

```
# Check the first rows  
head(str_df)
```

##	long	lat	stressor
## 1	-0.6628271	-3.4868198	Gillnet
## 2	4.4068230	0.2046171	Gillnet
## 3	1.3602443	-3.2101936	Gillnet
## 4	1.4402246	-6.5176442	Gillnet
## 5	2.2189534	-7.0911457	Gillnet
## 6	1.0967222	2.3932125	Gillnet

All we need now is to load the criteria table. This table compiles the key information required for the analysis, including exposure ratings of each species (or habitat) to specific stressors, consequence ratings of those interactions, and the corresponding weights used in risk estimation and the respective quality score of the data.

Since we are working within the *ByRA* framework, we will use the criteria table specifically designed for this purpose, following the approach described in Hines et al. (2020).

```
# Load criteria data  
criteria <- read.csv("data/criteria.csv")
```

```
# Check the first rows
head(criteria)
```

```
##              HABITAT.NAME Dolphin  X    X.1 Turtle X.2    X.3
## 1 HABITAT RESILIENCE ATTRIBUTES  RATING DQ WEIGHT RATING  DQ WEIGHT
## 2              age of maturity      3 3      2      3 3      2
## 3      reproductive strategy      3 3      3      1 3      3
## 4      population connectivity      2 3      2      1 3      2
## 5      local species status      2 3      2      3 3      2
## 6
##  CRITERIA.TYPE
## 1              E/C
## 2              C
## 3              C
## 4              C
## 5              C
## 6
##
## 1
## 2
## 3
## 4 (3) negligible movement/exchange between the focal regional population and other populations (a DP
## 5
## 6
```

Now we already have everything we need to run our first ByRA!

Let's start with a very simple example, where we just want a quick assessment of bycatch risk for two megafaunal species exposed to two different fishing gears. Since this is a first try, we will keep things straightforward and rely on the default behavior of the `quick_byra()` function.

This means we will not worry (yet) about the details behind how kernel density estimates for species and stressors are constructed, or how the HRA model is calculated. The default settings handle all of that for us, using conservative assumptions and standard data preparation protocols typically recommended for ByRA analyses.

We can always refine and tweak both the data prep and the HRA model later. So, for now, let's start simple and see what we get.

```
byra1 <- quick_byra(spp_df, str_df, criteria)
```

And that is it! All the results we want, including the maps and statistics of risk estimates are stored in `byra1`. If we check it out, we will find this:

```
names(byra1)
```

```
## [1] "Dolphin"          "Turtle"           "ecosys_risk_raw"
## [4] "ecosys_risk_classified" "summary_stats"    "kde_maps"
## [7] "area_of_interest"
```

Which means:

- **Dolphin and Turtle:** Partial results (risk estimates) for each species (can be just one or more!). This is a list containing maps of the Exposure and Consequence components of the risk assessment, as well as overall risk estimates for each individual stressor and their combined effects. It also includes a table with summary statistics for these estimates.
- **ecosys_risk_raw:** A `SpatRaster` map of the raw ecosystem risk estimates, representing the combined risk across all species considered in the assessment (Dolphin and Turtle).

- **ecosys_risk_classified:** A `SpatRaster` map of reclassified ecosystem risk estimates. ByRA typically uses four risk categories: 0 = None, 1 = Low, 2 = Medium, and 3 = High.
- **summary_stats:** A `data.frame` with summary statistics of risk estimates at both the ecosystem and species levels. It includes results for individual and combined stressor effects, as well as minimum, maximum, and mean values of Exposure, Consequence, and risk estimates, in addition to the relative area of each risk category.
- **kde_maps:** A list containing all kernel density estimate maps for each species and stressor, as well as maps of overlap for each species–stressor pair.
- **area_of_interest:** An `sf` vector object representing the polygon that defines the area of interest for the ByRA analysis. If not provided by the user, it is estimated by default based on the spatial distribution of the stressors.

First, let's check the summary stats!

```
byra1$summary_stats
```

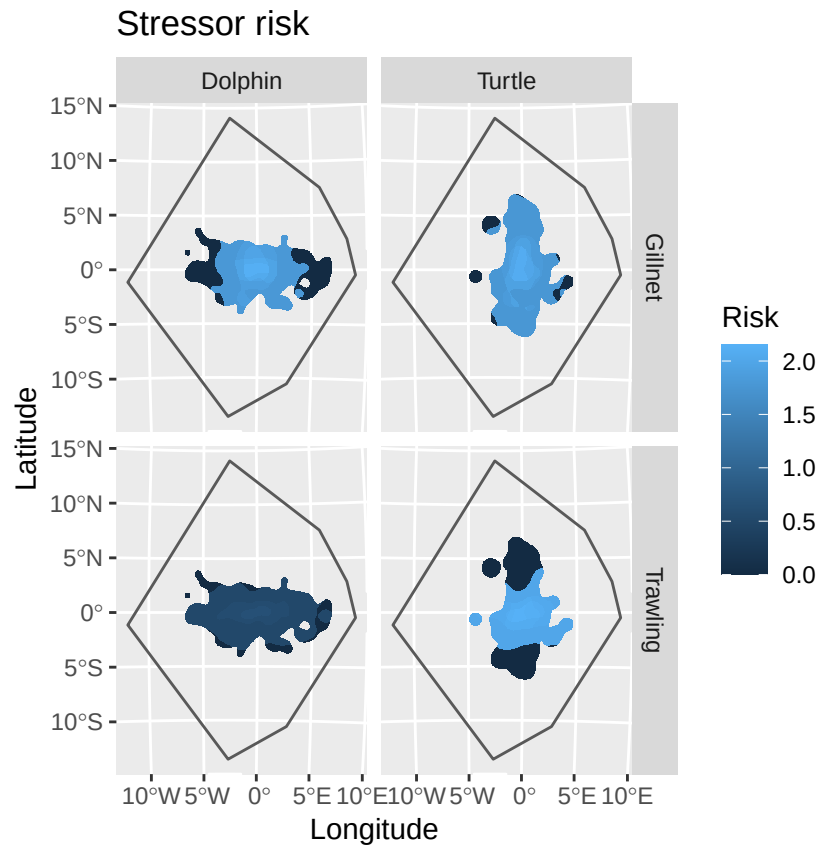
##	SPECIES	STRESSOR	E_min	E_max	E_mean	C_min	C_max	
## 3	ECOSYSTEM (FROM ALL STRESSORS)		NA	NA	NA	NA	NA	
## 1	Dolphin (FROM ALL STRESSORS)		0	2.470588	1.433456	0	2.529412	
## 2	Dolphin	Gillnet	0	2.470588	1.525953	0	2.529412	
## 31	Dolphin	Trawling	0	1.655172	1.340960	0	1.050000	
## 4	Turtle (FROM ALL STRESSORS)		0	2.647059	1.722917	0	2.750000	
## 5	Turtle	Gillnet	0	2.647059	2.234798	0	2.277778	
## 6	Turtle	Trawling	0	2.250000	1.211036	0	2.750000	
##	C_mean	R_min	R_max	R_mean	R.high	R.medium	R.low	R.None
## 3	NA	0	3.9206224	2.0025737	0.6722657	48.70748	42.64360	7.976654
## 1	1.2181434	0	2.1217281	0.8587702	0.0000000	63.23083	28.58950	8.179672
## 2	1.5891877	0	2.1217281	1.2503944	15.3867776	52.03508	0.00000	32.578143
## 31	0.8470992	0	0.6570775	0.4671461	0.0000000	0.00000	87.39600	12.604003
## 4	1.7201939	0	2.1505813	1.4052016	32.1657475	26.15591	38.00865	3.669697
## 5	1.9264752	0	2.0845908	1.6632733	19.6363094	71.85572	0.00000	8.507974
## 6	1.5139127	0	2.1505813	1.1471300	56.5628261	0.00000	0.00000	43.437174

Now let's visualize the main results using the `risaplot()` function. Since we are working with multiple species and stressors, the function will generate six plots.

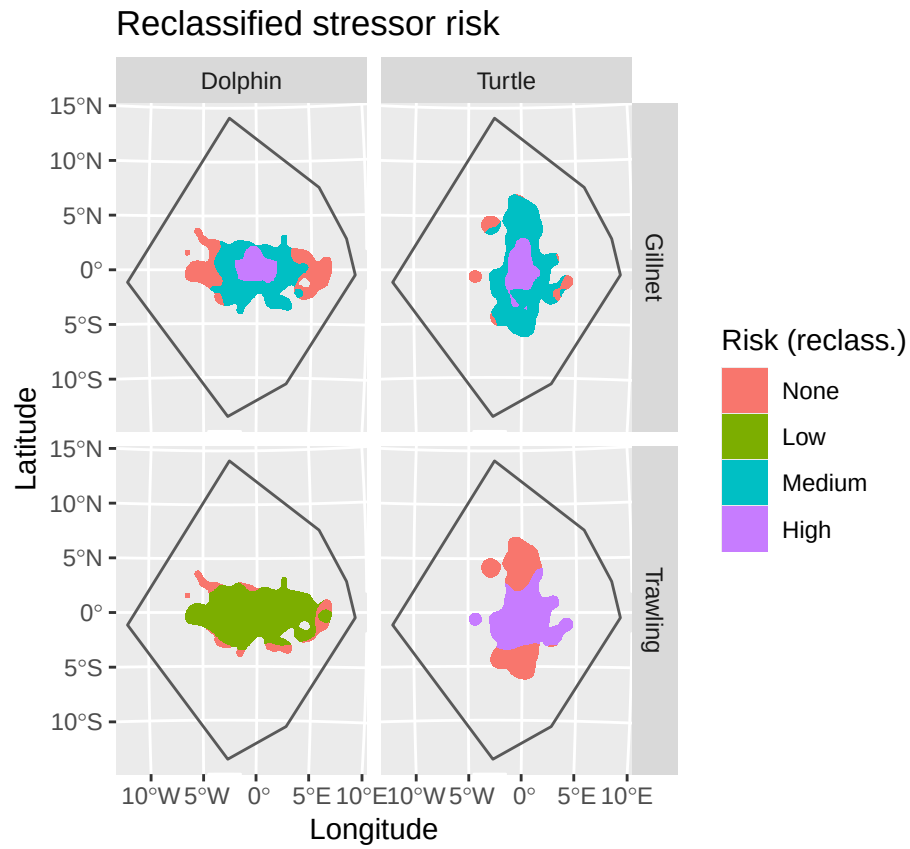
The first two plots show the risk estimates (raw and reclassified) for each species, considering the isolated effect of each stressor. Plots 3 and 4 display the risk estimates (raw and reclassified) for each species when all stressors are combined. Finally, plots 5 and 6 present the ecosystem-level risk estimates (raw and reclassified), derived from the combined risk across all species.

```
risaplot(byra1)
```

```
## [[1]]
```

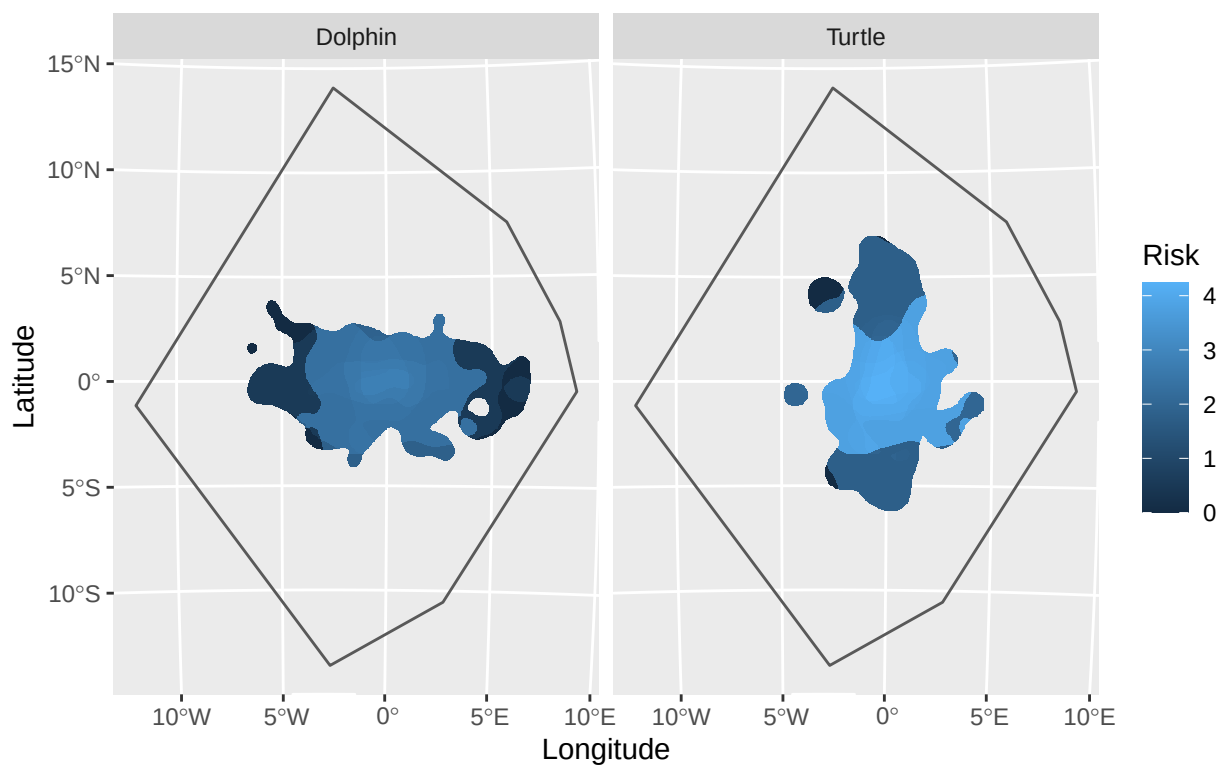


```
##
## [[2]]
```



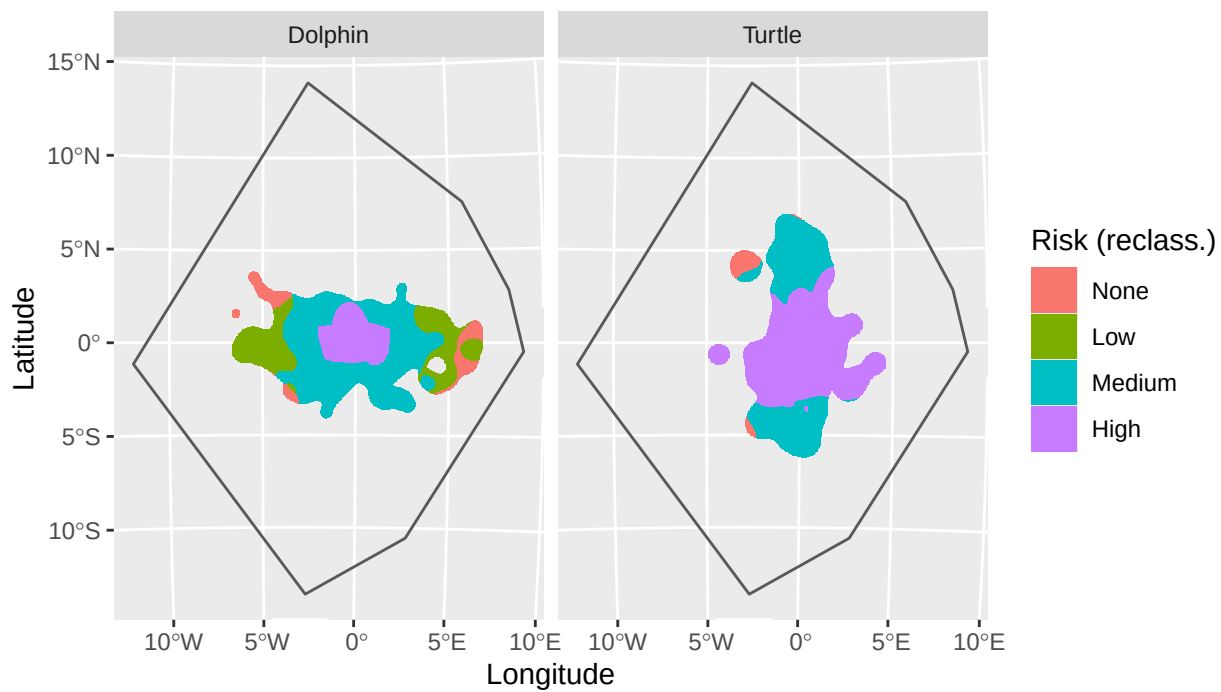
```
##  
## [[3]]
```

Total (combined stress.) risk

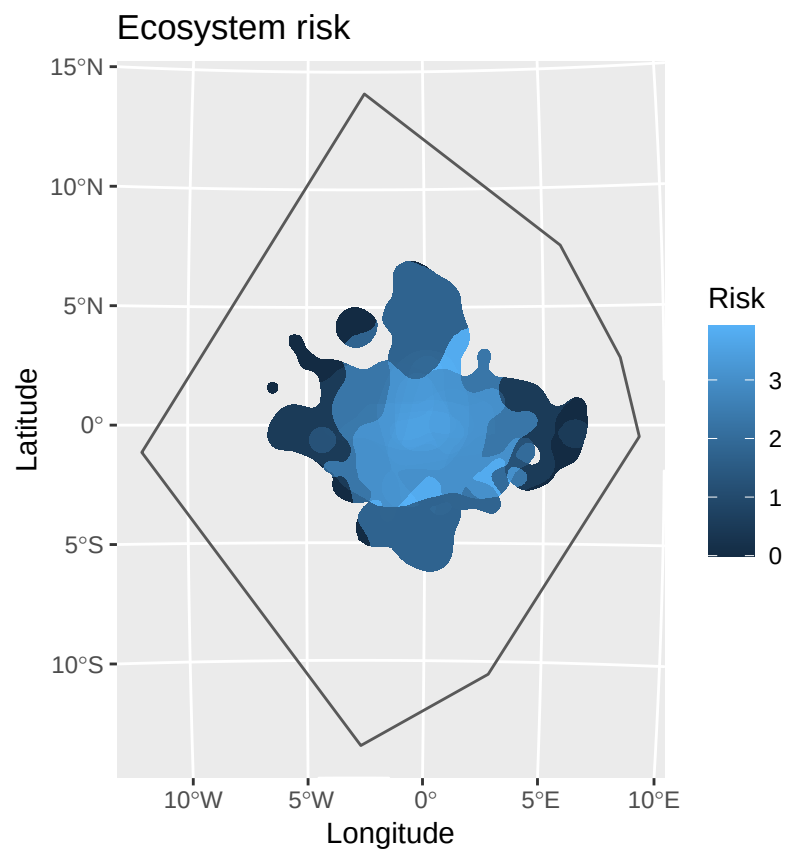


[[4]]

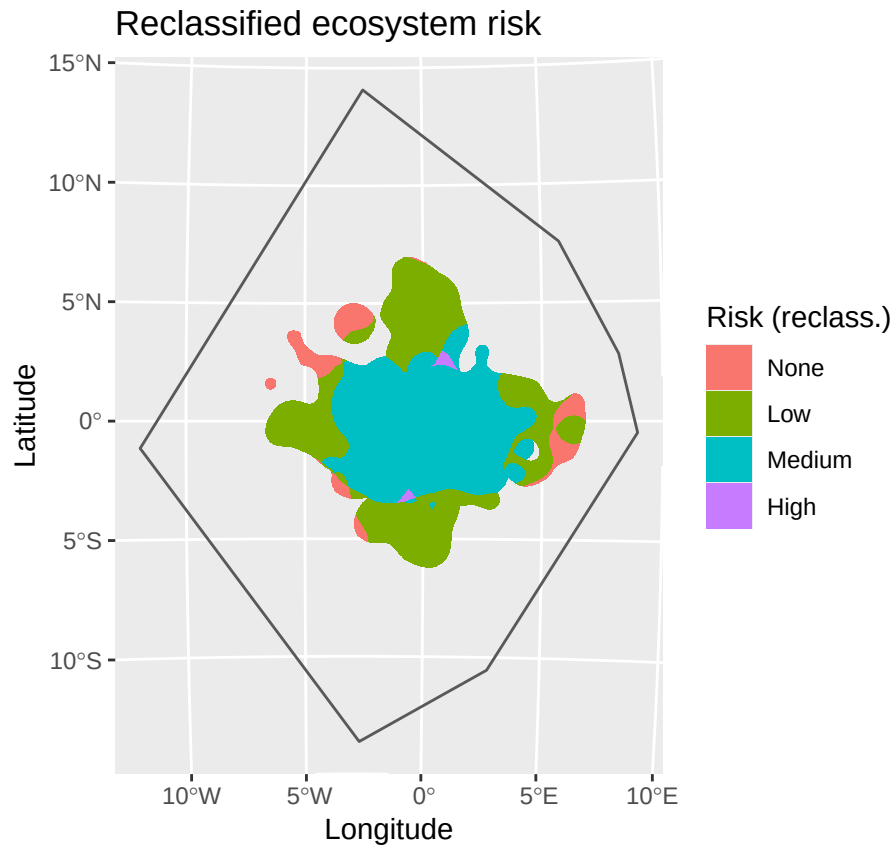
Reclassified total (combined stress.) risk



[[5]]



```
##  
## [[6]]
```

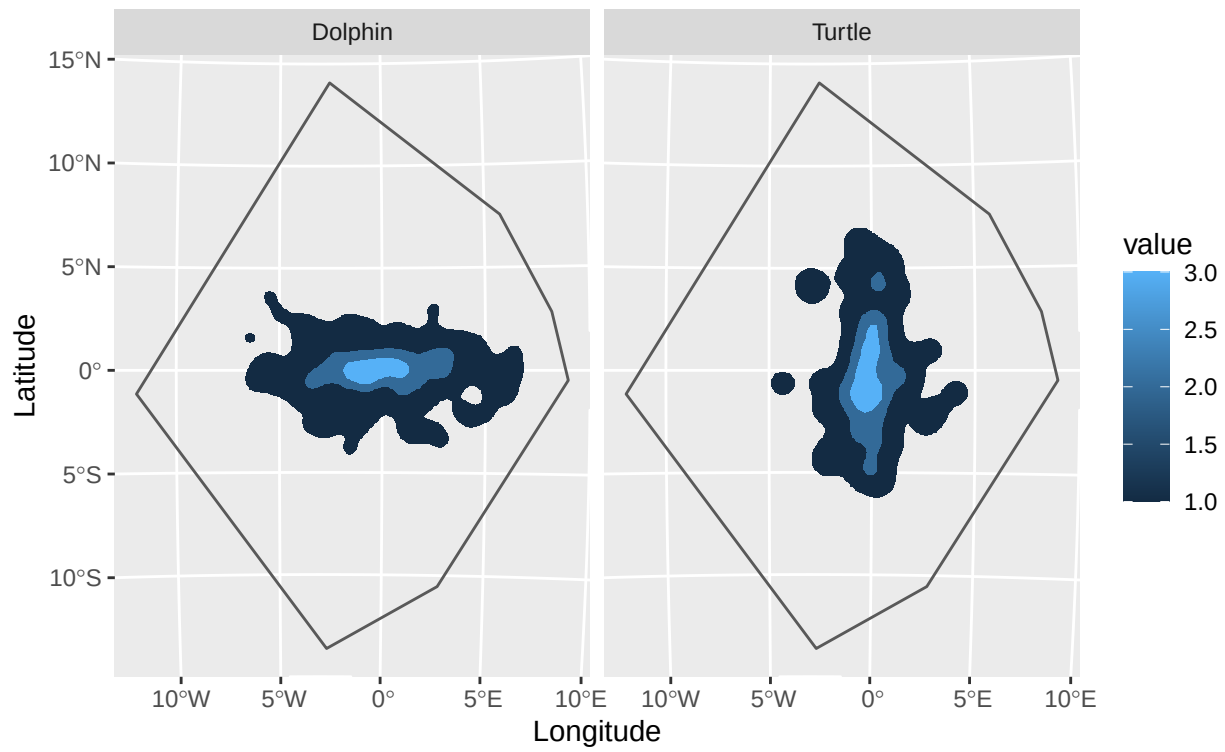



We can also visualize the maps that were considered for the risk estimates (KDE maps and their overlaps) using `risaplot()` applied to the `byra1$kde_maps` object. This will generate three plots: the first shows the KDE maps for each species, the second displays the KDE maps for each stressor, and the third presents the (reclassified) overlaps between each species–stressor pair.

```
risaplot(byra1$kde_maps)
```

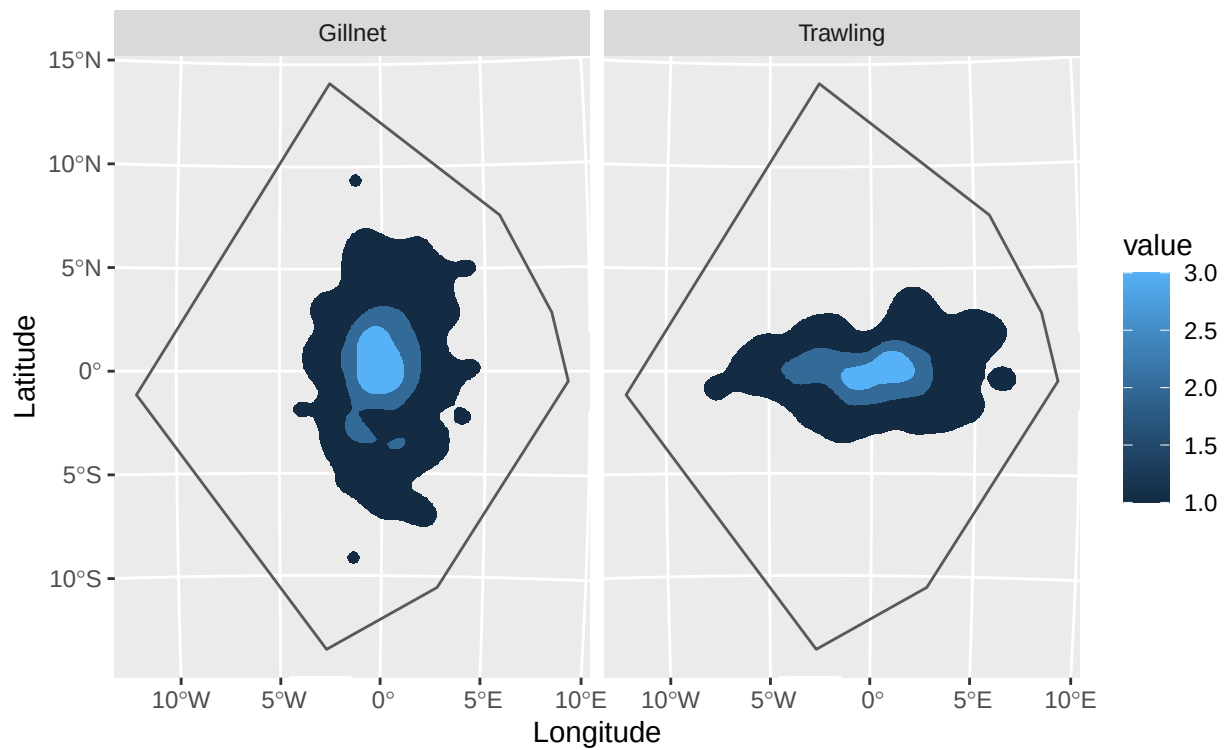
```
## [[1]]
```

Species KDE



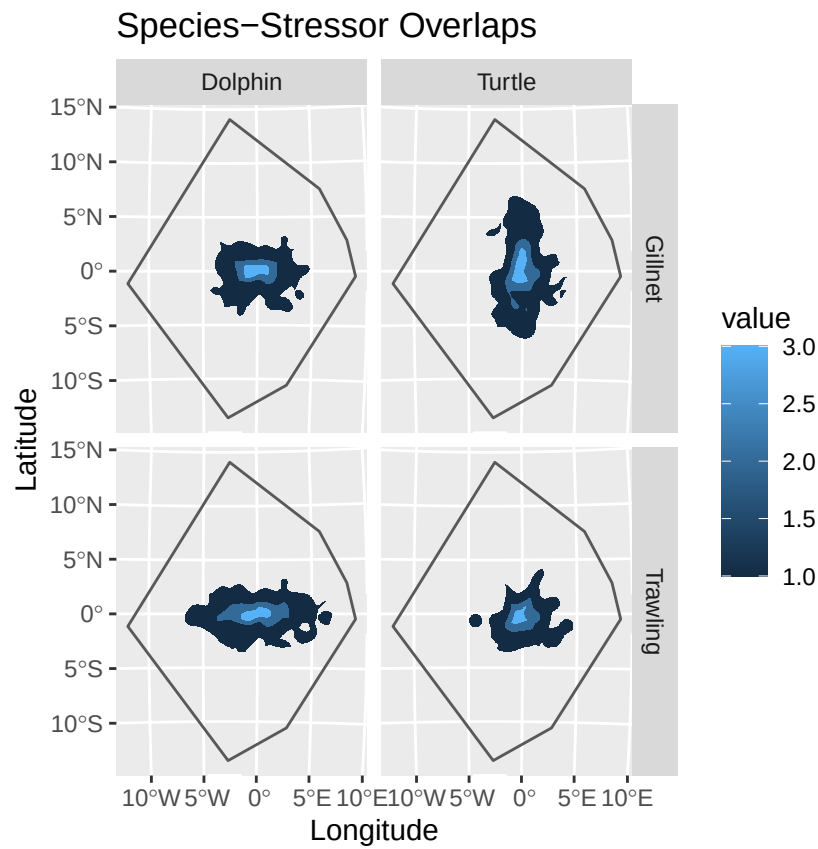
[[2]]

Stressor KDE



##

```
## [[3]]
```



References

Hines, EM, Johnson, A., Ponnampalam, L., Peter, C., Junchompoo, C., Vu, L., Thien, H., Caillat, M., and Verutes, G. (2020) Getting to the Bottom of Bycatch: A GIS-based toolbox to assess the risk of marine mammal bycatch. *Endangered Species Research* 42:37-57 DOI: <https://doi.org/10.3354/esr01037>